3/23/2022

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M168_Notes_7.4

7.4 Use of Counting Techniques in Probability

Let S be a uniform sample space and E be any event in S. Then

$$P(E) = \frac{\text{number of outcomes in event E}}{\text{number of outcomes in the Sample Space}} = \frac{n(E)}{n(S)}$$

Example

Suppose we have a jar with 8 blue and 6 green marbles. What is the probability that in a sample of 2, both will be blue?

$P(E) = \frac{4\times 7}{7\times 13} = \frac{4}{13}$

$\rightarrow C(14,2) = \frac{14!}{12!\times 2!} = 7\times 13 = n(S) = \frac{14!}{12!\times 2!} = \frac{14\times 13}{2} = 7\times 13$

$\rightarrow C(8,2) = \frac{8!}{6!\times 2!} = n(E) = 4\times 7$

$\frac{C(8,2)}{C(14,2)}$

What is the probability there is at least one blue marble?

$n(E) = \boxed{C(8,1)\times C(6,1)} + C(8,2)$

$= 8\times 6 + 4\times 7 = 48 + 28 = 76$

Find the probability distribution table for the number of blue marbles in the sample of 2 marbles:

Example

A stack of 100 copies has 3 defective papers. What is the probability that in a sample of 10 there will be no defective papers?

Example

A student takes a true/false test with 5 questions by guessing (choose answer at random). Write a probability distribution table for the number of correct answers.